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Surgical handle assembly

Abstract

The present disclosure includes apparatuses for a surgical handle assembly. An example apparatus includes a movable handle member and a switch configured to provide two or more modes of operation for the movable handle member and a driving pawl pivotally connected to a swing wheel configured to advance an actuation shaft linearly in a distal direction in response to actuation of the movable handle member.

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Background/Summary

PRIORITY INFORMATION

(1) This application claims benefit of U.S. Non-Provisional application Ser. No. 16/582,829 filed Sep. 25, 2019, and issued as U.S. Pat. No. 11,103,247 on Aug. 31, 2021, which claims the benefit of U.S. Non-Provisional application Ser. No. 15/481,949 filed Apr. 7, 2017 and issued as U.S. Pat. No. 10,433,842 on Oct. 8, 2019, the specifications of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates generally to a surgical handle assembly, and more particularly, to a surgical handle assembly configured to provide two or more modes of operation.

BACKGROUND

(3) A surgical handle assembly can be used in a number of surgical devices. One example includes use as a surgical stapler. A surgical stapler is a fastening device used to clamp tissue between opposing jaw structures to join tissue using surgical fasteners. Surgical staplers can include two elongated members used to clamp the tissue. One of the elongated members can include one or more reloadable cartridges and the other elongated member can include an anvil that can be used to form a staple when driven from the reloadable cartridge. A surgical stapler can receive one or more reloadable cartridges. An example of reloadable cartridges can include having rows of staples having a linear length. For example, a row of staples can have a linear length between 30 mm and

60 mm. A staple can be ejected by actuation of a movable handle member that is a part of the surgical handle assembly of the surgical stapler.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a surgical stapling apparatus including a surgical handle assembly in accordance with a number of embodiments of the present disclosure.
- (2) FIG. 2A is a schematic diagram of a surgical handle assembly in a first mode and a first movable handle member position in accordance with a number of embodiments of the present disclosure.
- (3) FIG. 2B is a schematic diagram of a surgical handle assembly in a first mode and a second movable handle member position in accordance with a number of embodiments of the present disclosure.
- (4) FIG. 3A is a schematic diagram of a surgical handle assembly in a second mode and a first movable handle member position in accordance with a number of embodiments of the present disclosure.
- (5) FIG. 3B is a schematic diagram of a surgical handle assembly in a second mode and a second movable handle member position in accordance with a number of embodiments of the present disclosure.
- (6) FIG. 4 is a schematic diagram of a surgical handle assembly including a first opening, a second opening, and a slot in accordance with a number of embodiments of the present disclosure.
- (7) FIG. 5 is a schematic diagram of a surgical handle assembly including a coupling point opening in accordance with a number of embodiments of the present disclosure.

DETAILED DESCRIPTION

- (8) The present disclosure includes apparatuses for a surgical handle assembly. An example apparatus includes a movable handle member and mode selection capability, e.g. a switch, configured to provide two or more modes of operation for the movable handle member. A driving pawl can be pivotally connected to a swing wheel to advance an actuation shaft linearly in a distal direction in response to actuation of the movable handle member.
- (9) In a number of embodiments, the mode selection capability can be a switch. The surgical handle assembly is described with the switch example throughout the present disclosure for ease of understanding and illustration; however embodiments are not limited to a switch. For example, the switch can include a first pin and a second pin to set modes of operation. While a first pin and a second pin are shown by example, more than two pins and/or two modes of operation are included in the embodiments disclosed herein. In this example, the first pin when set to a particular position can provide a first mode of operation for the movable handle member. The second pin when set to a particular position can provide a second mode of operation for the movable handle member. Use of the surgical handle assembly with a surgical stapler in the first mode of operation can advance an actuation shaft a first distance in the distal direction and coupled to a reloadable cartridge can deploy a first number of staples. Use of the surgical handle assembly with a surgical stapler in the second mode of operation can advance an actuation shaft a second distance in a distal direction and coupled to a reloadable cartridge can deploy a second, different number of staples. In a number of embodiments, the first mode of operation or the second mode of operation can be selected based on the type of tissue being fastened, a number of staples to be deployed, speed of staples being deployed, and/or an amount of force to be applied by a user to actuate the movable handle member, for example. A mode of operation can be selected based on the type of tissue being fastened, for example, when the tissue is thick, the user may use a mode that requires less force to be applied. A mode that requires less force also may be used when the user is fatigued or physically unable to use

a mode that requires more force, for example. A mode of operation can be selected based on speed of staples being deployed to lower edema, e.g. less blood in area being stapled, to allow stapling of tissue rather than blood.

(10) In a number of embodiments, the movable handle member may be coupled to a swing wheel. In this manner, the movable handle member and swing wheel may pivot around the first pin at a first pivot point during a first mode. The first pin can be connected to the movable handle member via a first opening in the swing wheel and movable handle member to engage the movable handle member. In a number of embodiments, the movable handle member and swing wheel may pivot around the second pin at a second pivot point during a second mode. The second pin can be connected to the movable handle member via a second opening in the swing wheel and movable handle member to engage the movable handle member. The second pin is not coupled to the movable handle member via the second opening while in the first mode of operation. The first pin is not coupled to the movable handle member via the first opening while in the second mode of operation.

(11) In a number of embodiments, transition from the first mode of operation to the second mode of operation can be accomplished by removing the first pin from the first opening while engaging the second pin with the second opening. The switch may be moved from a first position to a second position in order to engage and/or disengage the first and second pins with their respective openings. Transition from the second mode of operation to the first mode of operation can be accomplished by removing the second pin from the second opening while engaging the first pin with the first opening. Here, the switch may be moved from a second position to a first position to engage and/or disengage the first and second pins with their respective openings.

(12) In the following detailed description of the present disclosure, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration how one or more embodiments of the disclosure may be practiced. These embodiments are described in sufficient detail to enable those of ordinary skill in the art to practice the embodiments of this disclosure, and it is to be understood that other embodiments may be utilized and that process, electrical, and structural changes may be made without departing from the scope of the present disclosure.

(13) As used herein, designators such as “X”, “Y”, “N”, “M”, etc., particularly with respect to reference numerals in the drawings, indicate that a number of the particular feature so designated can be included. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. As used herein, the singular forms “a”, “an”, and “the” can include both singular and plural referents, unless the context clearly dictates otherwise. In addition, “a number of”, “at least one”, and “one or more” (e.g., a number of pivot points) can refer to one or more pivot points, whereas a “plurality of” is intended to refer to more than one of such things. Furthermore, the words “can” and “may” are used throughout this application in a permissive sense (i.e., having the potential to, being able to), not in a mandatory sense (i.e., must). The term “include,” and derivations thereof, means “including, but not limited to”. The terms “coupled” and “coupling” mean to be directly or indirectly connected physically or for access to and movement of the movable handle member, as appropriate to the context.

(14) The figures herein follow a numbering convention in which the first digit or digits correspond to the figure number and the remaining digits identify an element or component in the figure. Similar elements or components between different figures may be identified by the use of similar digits. For example, **210** may reference element “**10**” in FIG. 2A, and a similar element may be referenced as **310** in FIG. 3A. As will be appreciated, elements shown in the various embodiments herein can be added, exchanged, and/or eliminated so as to provide a number of additional embodiments of the present disclosure. In addition, the proportion and/or the relative scale of the elements provided in the figures are intended to illustrate certain embodiments of the present

disclosure and should not be taken in a limiting sense.

(15) FIG. 1 is a schematic diagram of a surgical stapling apparatus **102** including a surgical handle assembly **106** in accordance with a number of embodiments of the present disclosure. In the example, a surgical stapler apparatus **102** can include a surgical handle assembly **106** and a reloadable cartridge assembly **103**. As shown in the example of FIG. 1, the reloadable cartridge assembly **103**, e.g. a disposable loading unit, can be releasably secured to a distal end of an elongated body of the surgical handle assembly **106**. In this example, the reloadable cartridge assembly **103** can include a first elongated member **104** and a second elongated member **105** that can be used to clamp tissue. One of the elongated members can house one or more staple cartridges. The other elongated member can have an anvil that can be used to form a staple when driven from the staple cartridge. As mentioned, a surgical stapling apparatus **102** can receive reloadable cartridge assemblies having rows of staples. In a number of embodiments, third party reloadable cartridge and/or reloadable cartridge assemblies may be used with the surgical handle assembly **106** and embodiments of surgical handle assembly **106** may be configured to receive the same. A staple can be ejected by actuation of a movable handle member **110** that is a part of a surgical handle assembly **106** to the surgical stapling apparatus **102**. Actuation of the movable handle member **110** can actuate the actuation shaft (e.g. actuation shaft **240** in FIG. 2A) to eject a number of staples. For example, in a first mode of operation, actuation of the movable handle member **110** may eject **30** staples. In a second mode, actuation of the movable handle member **110** may eject **60** staples. However, embodiments are not limited to a particular number of staples ejected in the first or second mode. Further, embodiments are not limited to use with a surgical stapling apparatus. The surgical handle assembly **106** is described with the surgical stapling apparatus **102** example throughout the present disclosure for ease of understanding and illustration.

(16) FIG. 2A is a schematic diagram of a surgical handle assembly in a first mode and in a first movable handle member position (e.g., “ready” position) in accordance with a number of embodiments of the present disclosure. The surgical handle apparatus **206** can include a movable handle member **210**, a switch **220**, and a driving pawl **230**. The switch **220** can provide two or more modes of operation for the movable handle member **210**. The driving pawl **230** in a number of embodiments can be pivotally connected to a swing wheel **226** and can be configured to advance an actuation shaft **240** linearly in a distal direction in response to actuation of the movable handle member **210**.

(17) In a number of embodiments, the surgical handle assembly **206** can be in a first mode, as shown in FIG. 2A. The surgical handle assembly **206**, in a first mode, can have a first distance (e.g. distance **513** in FIG. 5) between a pivot point around first pin **222** to a coupling pin **232** of the driving pawl **230**. The surgical handle assembly **206**, in a second mode, can have a second distance (e.g. distance **515** in FIG. 5) between a pivot point around second pin **224** to the coupling pin **232** of the driving pawl **230**. In this example, the second distance (e.g. distance **515** in FIG. 5) is longer than the first distance (e.g. first distance **513** in FIG. 5). In a number of embodiments, a number of pivot points can exist and a pivot point can be around a pin placed in an opening in the movable handle **210**. In FIG. 2A, in the first mode, a first pin **222** can be coupled to the movable handle member **210** in a first opening and a second pin **224** can be disengaged from the second opening (e.g. second opening **454** in FIG. 4) so as not to be coupled to the movable handle member **210**. In the first mode the first pin **222** can serve as the pivot point for the movable handle member **210** with the first distance (e.g. distance **513** in FIG. 5) between the pivot point and coupling pin **232**. In a second mode, a second pin **224** can be coupled to the movable handle member **210**. In this example, the first pin **222** can be disengaged from the first opening (e.g. first opening **452** in FIG. 4) so as not to be coupled to the movable handle member **210**. In the second mode the second pin **224** can serve as the pivot point for the movable handle member **210**. In this example, the second distance (e.g. distance **515** in FIG. 5) is longer than the first distance (e.g. first distance **513** in FIG. 5). As such, in the first mode, when the first pin **222** is the pivot point, an amount of force used to

advance the actuation shaft **240** linearly in a distal direction is less than an amount of force used to advance the actuation shaft **240** linearly in the distal direction in a second mode of operation, using the second pin **224** as the pivot point. In this example, the amount of force used is less in the first mode than in the second mode due to the distance between the pivot point and the coupling pin **232** of the driving pawl **230** and because the travel distance is less.

(18) In a number of embodiments, the surgical handle assembly **206** can be in a first mode. In a first mode, the surgical handle assembly **206** can operate to advance the actuation shaft **240** a third distance (e.g. distance **544** in FIG. 5). The third distance (e.g. distance **544** in FIG. 5) is a length of advance of the actuation shaft **240** in a distal direction, upon actuation of the movable handle member **210** in the first mode. The surgical handle assembly **206** can alternatively be changed to a second mode of operation. In a second mode, the surgical handle assembly **206** can operate to advance the actuation shaft **240** a fourth distance (e.g. distance **546** in FIG. 5). The fourth distance (e.g. distance **546** in FIG. 5) is a length of advance of the actuation shaft **240** in the distal direction upon actuation of the movable handle member **210** in the second mode. In this example, the fourth distance (e.g. distance **546** in FIG. 5) is longer than the third distance (e.g. third distance **544** in FIG. 5).

(19) In a number of embodiments, the surgical handle assembly **206** can be in a ready position, as shown in FIG. 2A. In the ready position, the surgical handle assembly **206** is ready to advance the actuation shaft **240** linearly in a distal direction upon actuation of the movable handle member **210**.

(20) FIG. 2B is a schematic diagram of a surgical handle assembly in a first mode and a second movable handle member position (e.g. a “compressed” and/or “closed” position) in accordance with a number of embodiments of the present disclosure. The surgical handle assembly **206** can include a movable handle member **210**, a switch **220**, and a driving pawl **230**. The switch **220** can provide two or more modes of operation for the movable handle member **210**. The driving pawl **230** in a number of embodiments can be pivotally connected to a swing wheel **226** and can be configured to advance an actuation shaft **240** linearly in a distal direction in response to actuation of the movable handle member **210**.

(21) In a number of embodiments, the surgical handle assembly **206** can be in a first mode, as shown in FIG. 2B. The surgical handle assembly **206**, in a first mode, can have a first distance (e.g. distance **513** in FIG. 5) between the pivot point around first pin **222** to a coupling pin **232** of the driving pawl **230**. The surgical handle assembly **206**, in a second mode, can have a second distance (e.g. distance **515** in FIG. 5) from the pivot point around second pin **224** to the coupling pin **232** of the driving pawl **230**. In this example, the second distance (e.g. distance **515** in FIG. 5) is longer than the first distance (e.g. distance **513** in FIG. 5). In a number of embodiments, a number of pivot points can exist and a pivot point can be around a pin placed in an opening in the movable handle **210**. In FIG. 2B, in the first mode, the first pin **222** can be coupled to the movable handle member **210** in a first opening (e.g. first opening **452** in FIG. 4) and the second pin **224** can be disengaged from a second opening (e.g. second opening **454** in FIG. 4) so as not to be coupled to the movable handle member **210**. In the first mode, the first pin **222** can serve as the pivot point for the movable handle member **210** with the first distance (e.g. distance **513** in FIG. 5) between the pivot point and coupling pin **232**. In the second mode the second pin **224** can serve as the pivot point for the movable handle member **210**. In this example, the second distance (e.g. distance **515** in FIG. 5) is longer than the first distance (e.g. distance **513** in FIG. 5). Again, in the first mode, when the first pin **222** is the pivot point, an amount of force used to advance the actuation shaft **240** linearly in a distal direction is less than an amount of force used to advance the actuation shaft linearly in the distal direction in a second mode of operation, using the second pin **224** as the pivot point. In this example, the amount of force is less in the first mode than in the second mode due to the distance between the pivot point and the coupling pin **232** of the driving pawl **230** and the travel distance is less.

(22) In a first mode, the surgical handle assembly **206** can operate to advance the actuation shaft

240 a third distance (e.g. third distance **544** in FIG. 5). The third distance (e.g. distance **544** in FIG. 5) is a length of advance of the actuation shaft **240** in a distal direction, upon actuation of the movable handle member **210**. The surgical handle assembly **206** can be in a second mode. In a second mode, the surgical handle assembly **206** can operate to advance the actuation shaft **240** a fourth distance (e.g. distance **546** in FIG. 5). The fourth distance (e.g. distance **546** in FIG. 5) is a length of advance of the actuation shaft **240** in the distal direction. In this example, the fourth distance (e.g. distance **546** in FIG. 5) is longer than the third distance (e.g. distance **544** in FIG. 5). In a number of embodiments the fourth distance can be twice as long as the third distance. As such, the actuation shaft **240** can be advanced twice as far in the second mode than in the first mode

(23) In a number of embodiments, the surgical handle assembly **206** can be in a closed position, as shown in FIG. 2B. As will be explained more in connection with FIG. 4. In a closed position, the surgical handle assembly **206** can prevent the actuation shaft **240** from moving linearly in a distal and/or proximal direction upon release of the movable handle member **210** by having the driving pawl **230** disengage from teeth of a ratchet associated with the actuation shaft **240** and reengage with teeth further back at a starting point of the ratchet.

(24) In a number of embodiments, the pivot point of the surgical handle assembly **206** can be changed using the switch **220**. In an example, the switch **220**, when actuated, can rotate a gear **223**. The gear **223** can contact one or more gears of the first pin **222** and one or more gears of the second pin **224**. The surgical handle assembly **206** can be placed in the first mode when the switch **220** is actuated in a first direction. When the switch **220** is actuated in the first direction, the gear **223** can rotate in a first rotational direction. When the gear **223** is rotated in a first rotational direction the gear **223** can insert the first pin **222** into a first opening (e.g. first opening **452** in FIG. 4) and can remove the second pin **224** from a second opening (e.g. second opening **454** in FIG. 4). The surgical handle assembly **206** can be placed in the second mode when the switch **220** is actuated in a second direction. When the switch **220** is actuated in the second direction the gear can rotate in a second rotational direction. When the gear **223** is rotated in a second rotational direction the gear **223** can insert the second pin **224** into a second opening (e.g. second opening **454** in FIG. 4) and can remove the first pin **222** from the first opening (e.g. first opening **452** in FIG. 4). This example is used throughout the present disclosure for ease of understanding and illustration. Embodiments are not limited to a gear (e.g., gear **223** in FIG. 2A, for example) to transition between modes of operation. A number of mechanisms (e.g. a switch) to insert and remove a number of pins in a number of openings in a moveable handle member for operation in a number of modes can be used in accordance with a number of embodiments of the present disclosure to transition between modes of operation. For example, a push pin can be used to switch the surgical handle assembly **206** from one mode to another. A switch can be coupled to a first pin and a second pin such that when the first pin is pushed into the first opening (e.g. first opening **452** in FIG. 4) the second pin is removed from the second opening (e.g. second opening **454** in FIG. 4). As well as when the second pin is pushed into the second opening (e.g. second opening **454** in FIG. 4) the first pin is removed from the first opening (e.g. first opening **452** in FIG. 4).

(25) FIG. 3A is a schematic diagram of a surgical handle assembly in a second mode and a first movable handle position (e.g., “ready” position) in accordance with a number of embodiments of the present disclosure. The surgical handle assembly **306** can include a movable handle member **310**, a switch **320**, and a driving pawl **330**. The switch **320** can provide two or more modes of operation for the movable handle member **310**. The driving pawl **330** in a number of embodiments can be pivotally connected to the swing wheel **326** and can be configured to advance an actuation shaft **340** linearly in a distal direction in response to actuation of the movable handle member **310**.

(26) In a number of embodiments, the surgical handle assembly **306** can be in a second mode, as shown in FIG. 3A. The surgical handle assembly **306**, in a second mode, can have a second distance (e.g. distance **515** in FIG. 5) between the pivot point around a second pin **324** to the coupling pin **332** of the driving pawl **330**. In this example, the second distance is longer than the

first distance (e.g. distance **513** in FIG. 5). In a number of embodiments a number of pivot points can exist and a pivot point can be around a pin placed in an opening in the movable handle **310**. In FIG. 3A, in a second mode, a second pin **324** can be coupled to the movable handle member **310**. In this example, the first pin **322** can be disengaged from the first opening (e.g. first opening **452** in FIG. 4) so as not to be coupled to the movable handle member **310**. In the second mode the second pin **324** can serve as the pivot point for the movable handle member **310**. In a number of embodiments, the driving pawl **330** is configured to contact a toothed rack **342** of the actuation shaft **340** to advance the actuation shaft linearly in a distal direction. The actuation shaft **340** can be advanced in response to actuation of the movable handle member **310**. In this example, the second distance (e.g. distance **515** in FIG. 5) is longer than the first distance (e.g. first distance **513** in FIG. 5). As such, when the second pin **324** is the pivot point, the second mode operation advances twice as many ratchets of the toothed rack **342** than the first mode of operation. In this example, twice as many ratchets are advanced in the second mode due to the distance between the pivot point and the coupling pin **332** of the driving pawl **330**.

(27) In a number of embodiments, a swing wheel **326** is connected to the movable handle member **310**. Also, in a number of embodiments, the movable handle member can include a coupling opening (e.g. coupling opening **516** in FIG. 5). The swing wheel **326** and the coupling opening (e.g. coupling opening **516** in FIG. 5) can be used in the second mode of operation to allow transfer of force from the movable handle member **310** to the driving pawl **330**. Transferring the force from the movable handle member **310** to the driving pawl **330** allows the actuation shaft **340** to advance linearly in the distal direction.

(28) In a number of embodiments, the surgical handle assembly **306** can be in a ready position, as shown in FIG. 3A. In the ready position the surgical handle assembly **306** is ready to advance the actuation shaft **340** linearly in a distal direction upon actuation of the movable handle member **310**.

(29) FIG. 3B is a schematic diagram of a surgical handle assembly in a second mode and a second movable handle member position (e.g. a “compressed” and/or “closed” position) in accordance with a number of embodiments of the present disclosure. The surgical handle assembly **306** can include a movable handle member **310**, a switch **320**, and a driving pawl **330**. The switch **320** can provide two or more modes of operation for the movable handle member **310**. The driving pawl **330** in a number of embodiments can be pivotally connected to the swing wheel **326** and can be configured to advance an actuation shaft **340** linearly in a distal direction in response to actuation of the movable handle member **310**.

(30) In a number of embodiments, the surgical handle assembly **306** can be in a second mode, as shown in FIG. 3B. The surgical handle assembly **306**, in the second mode, can have a second distance (e.g. second distance **515** in FIG. 5) from the pivot point around the second pin **324** to the coupling pin **332** of the driving pawl **330**. In this example, the second distance (e.g. distance **515** in FIG. 5) is longer than the first distance (e.g. first distance **513** in FIG. 5). In a number of embodiments a number of pivot points can exist and a pivot point can be around a pin placed in an opening in the movable handle **310**. In a second mode, a second pin **324** can be coupled to the movable handle member **310**. In this example, the first pin **322** can be disengaged from the first opening (e.g. first opening **452** in FIG. 4) so as not to be coupled to the movable handle member **310**. In the second mode the second pin **324** can serve as the pivot point for the movable handle member **310**. In the second mode the second pin **324** can serve as the pivot point for the movable handle member **310**. In this example, the second distance (e.g. distance **515** in FIG. 5) is longer than the first distance (e.g. distance **513** in FIG. 5). In a number of embodiments, the driving pawl **330** is configured to contact a toothed rack **342** of an actuation shaft **340** to advance the actuation shaft linearly in a distal direction in response to actuation of the movable handle member **310**. In this example, the second distance (e.g. distance **515** in FIG. 5) is longer than the first distance (e.g. first distance **513** in FIG. 5). As such, when the second pin **324** serves as the pivot point, the second mode of operation advances the actuation shaft **340** twice as far and advances twice as many

ratchets of the toothed rack than the first mode of operation. In this example, the actuation shaft **340** is advanced twice as far and twice as many ratchets are advanced in the second mode due to the distance between the pivot point and the coupling pin **332** of the driving pawl **330**.

(31) In a number of embodiments, the surgical handle assembly **306** can be in a closed position, as shown in FIG. **3B**. In a closed position, the surgical handle assembly **306** can prevent the actuation shaft **340** from moving linearly in a distal and/or proximal direction upon release of the movable handle member **310**.

(32) FIG. **4** is a schematic diagram of a surgical handle assembly including a first opening, a second opening, and a slot in accordance with a number of embodiments of the present disclosure. In FIG. **4**, the movable handle member **410** can include a first opening **452** and a second opening **454**. The first pin (e.g. first pin **222** in FIG. **2A**) can be inserted into the first opening **452** for the first mode. The second pin (e.g. second pin **224** in FIG. **2A**) can be inserted into the second opening **454** for the second mode. In a number of embodiments, the swing wheel **426** can include a slot **428**. The slot **428** can allow the swing wheel **426** to rotate whether the first pin (e.g. first pin **222** in FIG. **2A**) is inserted into the first opening **452** or the first pin (e.g. first pin **222** in FIG. **2A**) is removed from the first opening **452**.

(33) FIG. **5** is a schematic diagram of a surgical handle assembly including a coupling opening **516** in accordance with a number of embodiments of the present disclosure. In a number of embodiments, the movable handle member **510** can include a coupling opening **516**. The coupling pin (e.g. coupling pin **232** in FIG. **2A**) can be pivotally connected to the coupling opening **516**. The coupling opening **516** can allow the actuation shaft **540** to advance linearly in the distal direction without coupling pin **232** binding due to the shape of the coupling opening **516**. The coupling opening **516** shape, for example, can be a “U” shape, e.g. saddle, allowing degrees of translation and movement in contrast to being a fixed point.

(34) In a number of embodiments, the surgical handle assembly **506** can be in a first mode. In the first mode, the first pin (e.g. first pin **222** in FIG. **2A**) serves as the pivot point, having a first distance **513**. In a second mode, the second pin (e.g. second pin **224** in FIG. **2A**) serves as the pivot point, having a second distance **515**. In this example, the second distance **515** is longer than the first distance **513**.

(35) In a number of embodiments, the surgical handle assembly **506** can be in a first mode having a third distance **544**. The third distance **544** is a length of advance of the actuation shaft **540** in the distal direction. The surgical handle assembly **506** can be in a second mode having a fourth distance **546**. The fourth distance **546** is a length of advance of the actuation shaft **540** in the distal direction. In this example, the fourth distance **546** is longer than the third distance **544**.

(36) FIG. **5** illustrates the surgical handle assembly **506** in a second mode and the movable handle member **510** in a closed position. The driving pawl (e.g. driving pawl **230** in FIG. **2A**), in this example, is at the end of the fourth distance **546** and has fully advanced the actuation shaft **540** for an actuation of the movable handle member **510** in the second mode of operation. In a first mode and when the movable handle member **510** is in a closed position, the driving pawl (e.g. driving pawl **230** in FIG. **2A**) can be at the end of the third distance **544** and has fully advanced the actuation shaft **540** for an actuation of the movable handle member **510** in the first mode of operation.

(37) Although specific embodiments have been illustrated and described herein, those of ordinary skill in the art will appreciate that an arrangement calculated to achieve the same results can be substituted for the specific embodiments shown. This disclosure is intended to cover adaptations or variations of one or more embodiments of the present disclosure. It is to be understood that the above description has been made in an illustrative fashion, and not a restrictive one. Combination of the above embodiments, and other embodiments not specifically described herein will be apparent to those of skill in the art upon reviewing the above description. The scope of the one or more embodiments of the present disclosure includes other applications in which the above

structures and processes are used. Therefore, the scope of one or more embodiments of the present disclosure should be determined with reference to the appended claims, along with the full range of equivalents to which such claims are entitled.

(38) In the foregoing Detailed Description, some features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the disclosed embodiments of the present disclosure have to use more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

Claims

1. A surgical handle assembly apparatus, comprising: a movable handle member; an actuation shaft; and a switch configured to: (a) provide a first mode of operation for the movable handle member by connecting a first pin to the movable handle member via a first opening and (b) provide a second mode of operation for the movable handle member by connecting a second pin to the movable handle member via a second opening, wherein, in the first mode of operation, the actuation shaft is advanced linearly in a distal direction a first distance in response to actuation of the movable handle member, wherein, in the second mode of operation, the actuation shaft is advanced linearly in the distal direction a second distance in response to actuation of the movable handle member, and wherein the first distance is different from the second distance.
2. The apparatus of claim 1, wherein: in the first mode of operation, a first amount of force is required to move the movable handle; in the second mode of operation, a second amount of force is required to move the movable handle; and the first amount of force is different from the second amount of force.
3. The apparatus of claim 1, further comprising a driving pawl configured to advance the actuation shaft.
4. The apparatus of claim 3, further comprising a swing wheel, wherein the driving pawl is pivotally connected to the swing wheel, and wherein the swing wheel is connected to the movable handle.
5. The apparatus of claim 1, wherein the switch includes the first pin and the second pin.
6. A surgical handle assembly apparatus, comprising: a movable handle member; a switch configured to: (a) provide a first mode of operation for the movable handle member by connecting a first pin to the movable handle member via a first opening and (b) provide a second mode of operation for the movable handle member by connecting a second pin to the movable handle member via a second opening, wherein, in the first mode of operation, the movable handle pivots on a first pivot point, and wherein, in the second mode of operation, the movable handle pivots on a second pivot point.
7. The apparatus of claim 6, further comprising an actuation shaft, wherein the actuation shaft is advanced linearly in a distal direction a first distance in response to actuation of the movable handle member during the first mode of operation, and wherein the actuation shaft is advanced linearly in the distal direction a second distance in response to actuation of the movable handle member during the second mode of operation.
8. The apparatus of claim 6, wherein: in the first mode of operation, a first amount of force is required to move the movable handle; in the second mode of operation, a second amount of force is required to move the movable handle; and the first amount of force is different from the second amount of force.
9. The apparatus of claim 6, wherein the switch includes the first pin and the second pin.
10. The apparatus of claim 6, further comprising a swing wheel connected to the movable handle.

11. The apparatus of claim 6, wherein the movable handle member pivots around the first pin at the first pivot point during the first mode of operation, and wherein the movable handle member pivots around the second pin at the second pivot point during the second mode of operation.
 12. The apparatus of claim 6, further comprising a driving pawl, wherein: in the first and second modes of operation, the driving pawl is configured to: advance an actuation shaft linearly in a distal direction to deploy one or more staples, and the apparatus is configured to deploys a different number of staples in the first mode of operation than in the second mode of operation.
 13. A surgical handle assembly apparatus, comprising: a movable handle member; and a switch configured to (a) provide a first mode of operation for the movable handle member by connecting a first pin to the movable handle member via a first opening and (b) to provide a second mode of operation for the movable handle member by connecting a second pin to the movable handle member via a second opening, wherein, in the first mode of operation, a first amount of force is required to move the movable handle, wherein, in the second mode of operation, a second amount of force is required to move the movable handle, and wherein the first amount of force is different from the second amount of force.
 14. The apparatus of claim 13, further comprising a staple cartridge, wherein: a driving pawl is configured to: advance a toothed rack linearly in a distal direction in response to actuation of the movable handle to deploy one or more staples in the first and second modes of operation, and the apparatus is configured to deploys a different number of staples in the first mode of operation than in the second mode of operation.
 15. The apparatus of claim 14, wherein the one or more staples are contained in a staple cartridge removably connected to the toothed rack.
 16. The apparatus of claim 13, wherein a driving pawl is configured to: (i) advance a toothed rack linearly in a distal direction a first distance in response to actuation of the movable handle member during the first mode of operation and (ii) advance the toothed rack linearly in the distal direction a second distance in response to actuation of the movable handle member during the second mode of operation.
 17. The apparatus of claim 13, further comprising a swing wheel connected to the movable handle member and to a driving pawl.
 18. The apparatus of claim 13, wherein the first pin is not coupled to the movable handle member via the first opening during the second mode of operation.
 19. The apparatus of claim 13, wherein the second pin is not coupled to the movable handle member via the second opening during the first mode of operation.
 20. The apparatus of claim 13 wherein the switch includes the first pin and the second pin.
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